

Curriculum Vitae of J. Miguel Valverde

Personal details

👤 Full name: Juan Miguel Valverde Martinez
🌐 Website: <https://jmlipman.github.io/>
📖 Google scholar: <https://scholar.google.com/citations?user=OPH9yeIAAAAJ&hl=en>
🆔 ORCID: <http://orcid.org/0000-0002-2708-1547>
🐙 Github: <https://github.com/jmlipman>

Work experience

DTU Compute, Technical University of Denmark Lyngby, Denmark 🇩🇰
Postdoc Researcher
Sep 2023–Aug 2026
Development of topology- and morphology-aware deep learning segmentation methods.
Supervisor: Anders Bjorholm Dahl, Rasmus Larsen

A.I. Virtanen Institute, University of Eastern Finland Kuopio, Finland 🇫🇮
Postdoc Researcher
Oct 2022–Aug 2023
Segmentation of electron microscopy images with deep learning.
Supervisor: Alejandra Sierra

Department of Architecture, Sojo University Kumamoto, Japan 🇯🇵
Researcher
Oct 2017–Sep 2018
Navigation map system for wheelchair users for iOS (front and backend).
Supervisor: Motoya Koga

Copenhagen Business School Copenhagen, Denmark 🇩🇰
Research student
Dec 2015–Mar 2016
Visualizing Public Complex Data.

Education

University of Eastern Finland Kuopio, Finland 🇫🇮
Ph.D. in Computer Science (with honors)
Oct 2018–Sep 2022
Thesis: MRI Segmentation with Deep Learning. Application in Rodent Brain Images with Lesions [🔗](#) (examiner: Koen Van Leemput). Supervisors: Jussi Tohka, Olli Gröhn

University of Oxford (*Research visit*) Oxford, UK 🇬🇧
Supervisor: Jason Lerch
Nov 2021–Feb 2022

University of Copenhagen Copenhagen, Denmark 🇩🇰
M.Sc. in IT and Cognition
Aug 2015–Sep 2017
Thesis: Template building and labelling of in-vivo specimens of rats with hypertension in pre-clinical settings. Supervisor: Sune Darkner

University of Burgos Burgos, Spain 🇪🇸
BSc. in Computer Science Engineering (*Grado en Ingeniería informática*)
Sep 2010–Jun 2014

Turku University of Applied Sciences (*Erasmus Exchange*) Turku, Finland 🇫🇮
Aug 2013–Jun 2014

Awards, Mentions, Distinctions

BMVC Outstanding Reviewer 🔗	2025
CVPR Outstanding Reviewer 🔗	2025
NeurIPS Top Reviewer 🔗	2024
MICCAI Outstanding Reviewer – Honorable mention 🔗	2023
UEF's Young Researcher Award 🔗	2023
Ph.D. Thesis and defense – Approved with honors 🔗	2022
MICCAI Outstanding Reviewer – Honorable mention 🔗	2022

Funding

Technical University of Denmark 4.43M DKK
HAV-INNO Project, with Jon Svendsen, and Malte Pedersen Sep 2026-Dec 2028

University of Eastern Finland 550€
For the research visit to the University of Oxford Jan 2022

Otto Malm Foundation 4300€ For the research visit to the University of Oxford	Jun 2020
University of Eastern Finland 900€ For attending MICCAI conference in Shenzhen, China	Sep 2019

Teaching

Introduction to Deep Learning in Computer Vision <i>Technical University of Denmark</i> with Aasa Feragen, Dimitrios Papadopoulos	Lyngby, Denmark 2025
Deep Learning for Experimental 3D Image Analysis <i>Technical University of Denmark</i> with Hans Martin Kjer, Marco Pizzolato, Vedrana A. Dahl, Anders B. Dahl	Lyngby, Denmark 2024, 2025

Supervision

Killian Victor Véronique Legrand <i>Swiss Federal Technology Institute of Lausanne (EPFL)</i> MSc. Thesis: “Advanced Segmentation Methods for Robust Mineral Classification in Multi-Modal X-ray and Neutron Tomography”	Lyngby, Denmark Feb 2026-Jun 2026
Paula Barho <i>Technical University of Denmark</i> MSc. Thesis: “Synthetic Data Generation for Polyp Segmentation in Limited-Label Endoscopic Imaging”	Lyngby, Denmark Feb 2026-Jun 2026
Sólón Núi Sindrasón & David Domingo Hansson <i>Technical University of Denmark</i> MSc. Thesis: “Augmented-reality (AR) headsets for on-site maintenance of drones using computer vision”	Lyngby, Denmark Feb 2026-Jun 2026
Thor-Haakon Howatson Wichmann Jørgensen <i>Technical University of Denmark</i> MSc. Thesis: “3D Instance Segmentation of Axons in microscopic Images with Deep Learning”	Lyngby, Denmark Feb 2026-Jun 2026
Samer Bujana Escalona <i>Technical University of Denmark</i> MSc. Thesis: “Neural Network-Enhanced Material Recognition Using NIR Polarization Imaging”	Lyngby, Denmark Feb 2026-Jun 2026
Enrico Tuda <i>Technical University of Denmark</i> MSc. Thesis: “Object Detection of Marine Engine Components with Deep Learning Using Synthetic Data from CAD Models”	Lyngby, Denmark Oct 2025-Mar 2026
Silas Krongaard Hansen & Agnes Lund Olsen <i>Technical University of Denmark</i> BSc. Thesis: “Segmentation and Characterization of Plant Root System Architecture from Micro-CT Scans”	Lyngby, Denmark Feb 2025-Jun 2025
Julius Ellegård Grønager <i>Technical University of Denmark</i> BSc. Thesis: “Improving Skeleton Consistency in 3D Medical Image Segmentations of Tubular Structures Using Deep Learning and Visual Analytics”	Lyngby, Denmark Jan 2025-Jun 2025
Morten Møller Christensen <i>Technical University of Denmark</i> MSc. Thesis: “Applying machine learning approaches for estimating temporal changes in surface water extent using multi-modal remote sensing data”	Lyngby, Denmark Jan 2025-Jun 2025
Carl Lucas Grundtvig Nørlund & Malthe August Bordin Bresler <i>Technical University of Denmark</i> BSc. Thesis: “Thin Object Image Segmentation in Three Dimensions”	Lyngby, Denmark Jan 2024-Jun 2024
Mohammad Jaber Hossain <i>Norwegian University of Science and Technology</i> MSc. Thesis: “Transfer Learning-Based Deep Learning Techniques for Brain Lesion Segmentation in 3D Pre-Clinical Magnetic Resonance Images”	Kuopio, Finland Jan 2023-Aug 2023
Sergio Doval Sánchez <i>Universidad Carlos III de Madrid</i> BSc. Thesis: “Automated Segmentation of Ischemic Lesions in MR Images”	Kuopio, Finland Sep 2022-Jun 2023

Other key academic merits

Organization, Staff

ECCV Workshop on Computer Vision for Natural Heritage <i>Organizing committee</i> Roberta E. Hunt, H. Martin Kjer, Anders B. Dahl, and Kim Steenstrup Petersen.	Malmö, Sweden September 2026
Multi-modal Anomaly Detection Workshop (MMAD) <i>Co-organizer</i> with Kevin Wilkinghoff, Neelu Madan.	Aalborg, Denmark February 2026
Scandinavian Conference on Image Analysis (SCIA) <i>Organizing committee</i> with Rasmus Larsen, Thomas B. Moeslund, Jens Petersen, Vedrana A. Dahl.	Reykjavik, Iceland June 2025
Workshop on Challenges in analysis of large image data <i>Organizing committee</i> with Anders B. Dahl, Rebecca Engberg, and the rest of QIM Centre.	Reykjavik, Iceland June 2025
Navigating as a Researcher in the Era of AI (Workshop) <i>Debate coach</i> with Gretchen Repasky, Benjamin Lebiecka-Johansen, Katharina Herzog, Tibor Varga.	Aalborg, Denmark March 2024
UEF Winter School <i>Award committee</i> with Jenni Küblbeck (UEF, main organizer).	Kuopio, Finland May 2023
MLT <code>__init__</code> journal club <i>Co-organizer</i> with Jayson Cunanan (AI Inside), Suzana Ilić (Hugging Face), Alisher Abdulkhaev (ROMS).	 Online Jan 2021-Sep 2022
UEF Winter School <i>Co-organizer</i> with Joanna Huttunen, Lenka Dvoráková, Ratika Sehgal, Heidi Pulkkinen, Niina Vuokila, Sami Gabbouj.	Kuopio, Finland Mar 2019
Nordic AI <i>Volunteer</i> with Sarah Gill (Founders).	Copenhagen, Denmark Sep 2017

Invited lectures

Board of European Students of Technology (BEST) <i>Lecture</i> Topic: Image recognition & AI	Lyngby, Denmark June 2025
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International research activity

Integrating AI & Data Sharing for the Human Organ Atlas Hub workshop <i>Poster presentation</i> Disconnect to Connect: A Data Augmentation Method for Improving Topology Accuracy in Image Segmentation	Lyngby, Denmark Nov 2024
MAX IV User's Meeting <i>Poster presentation</i> Disconnect to Connect: A Data Augmentation Method for Improving Topology Accuracy in Image Segmentation	Lund, Sweden Jan 2024
Nordic Probabilistic AI school <i>Poster presentation</i> Region-wise Loss for Biomedical Image Segmentation	Helsinki, Finland Jun 2022
Challenge in Adolescent Brain Cognitive Development Neurocognitive Prediction at MICCAI Conference <i>Poster presentation</i> Predicting intelligence based on cortical WM/GM contrast, cortical thickness and volumetry	Shenzhen, China Oct 2019
International Workshop on Machine Learning in Medical Imaging at MICCAI Conference <i>Poster presentation</i> Automatic Rodent Brain MRI Lesion Segmentation with Fully Convolutional Networks	Shenzhen, China Oct 2019
Machine Learning Summer School <i>Poster presentation</i> Automatic Rodent Brain MRI Lesion Segmentation with Fully Convolutional Networks	Moscow, Russia Aug 2019
University of Eastern Finland Winter School <i>Poster presentation</i> Automatic Rodent Brain MRI Lesion Segmentation with Fully Convolutional Networks	Kuopio, Finland Mar 2019

Peer review

CVPR	2025-2026
ECCV	2026

NeurIPS	2024-2026
BMVC	2025-2026
MICCAI International Conference	2022-2026
npj Digital Medicine	2026
IEEE Journal of Biomedical and Health Informatics	2026
WACV	2025
AI in Science Summit (AIS)	2025
TMLR	2025
ICLR	2025
Communications Medicine	2024
Entropy	2024
Scientific Reports	2022,2024
MICAD International Conference	2023
Quantitative Imaging in Medicine and Surgery	2023
Artificial Intelligence In Medicine	2023
Frontiers in Neuroimaging	2022
Frontiers in Neurology	2022
BMC Medical Imaging	2021-2022
Biocybernetics and Biomedical Engineering	2021
Computational Intelligence and Neuroscience	2021

Language skills

-  Spanish: Native
-  English: Full professional proficiency
-  Japanese: Elementary proficiency

Publications

 Open access  Preprint  Code * Shared authorship

Peer-reviewed Conference Articles

1. J.M. Valverde, D.P. Papadopoulos, R. Larsen, A.B. Dahl. **Towards High-Quality Image Segmentation: Improving Topology Accuracy by Penalizing Neighbor Pixels.** In *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.  
2. J.M. Valverde, M. Koga, N. Otsuka, A.B. Dahl. **TopoMortar: A dataset to evaluate image segmentation methods focused on topology accuracy.** In *British Machine Vision Conference (BMVC) (Oral)*, 2025.  
3. J.M. Valverde, A. Shatillo, R. De Feo, O. Gröhn, A. Sierra, J. Tohka. **Automatic Rodent Brain MRI Lesion Segmentation with Fully Convolutional Networks.** In *International Workshop on Machine Learning in Medical Imaging* (pp. 195-202). Springer, Cham. 2019  
4. J.M. Valverde*, V. Imani*, J.D. Lewis, J. Tohka. **Predicting intelligence based on cortical WM/GM contrast, cortical thickness and volumetry.** In *Challenge in Adolescent Brain Cognitive Development Neurocognitive Prediction* (pp. 57-65). Springer, Cham. 2019  

Peer-reviewed Journal Articles

1. J.M. Valverde, V. Imani, J. Tohka. **Fine-tuning ImageNet-pretrained Models in Medical Image Classification: Reassessing the Impact of Confounding Factors.** In *Pattern Recognition Letters*, 2026  
2. J.M. Valverde, A. Shatillo, J. Tohka. **Sauron U-Net: Simple automated redundancy elimination in medical image segmentation via filter pruning.** In *Neurocomputing*, 2024   
3. A. Lazari, M. Tachrount, J.M. Valverde, D. Papp, A. Beauchamp, P. McCarthy, J. Ellegood, J. Grandjean, H. Johansen-Berg, V. Zerbi, J. P Lerch, R. B Mars **The mouse motor system contains multiple premotor areas and partially follows human organizational principles.** In *Cell Reports*, 2024  
4. J.M. Valverde, J. Tohka. **Region-wise Loss for Biomedical Image Segmentation.** In *Pattern Recognition*, 2023   

5. [J.M. Valverde](#), A. Shatillo, R. De Feo, J. Tohka. **Automatic cerebral hemisphere segmentation in rat MRI with lesions via attention-based convolutional neural networks.** In *NeuroInformatics*, 2022    
6. R. De Feo, E. Hämäläinen, E. Manninen, R. Immonen, [J.M. Valverde](#), X.E. Nnode-Ekane, O. Gröhn, A. Pitkänen, J. Tohka. **Convolutional Neural Networks Enable Robust Automatic Segmentation of the Rat Hippocampus in MRI After Traumatic Brain Injury.** In *Frontiers in Neurology*, 2022    
7. [J.M. Valverde](#), V. Imani, A. Abdollahzadeh, R. De Feo, M. Prakash, R. Ciszek, J. Tohka. **Transfer Learning in Magnetic Resonance Brain Imaging: A Systematic Review.** In *Journal of Imaging*, 2021   
8. R. De Feo, A. Shatillo, A. Sierra, [J.M. Valverde](#), O. Gröhn, J. Tohka. **Automated skull-stripping and segmentation with Multi-Task U-Net in large mouse brain MRI databases.** In *NeuroImage*, 2021    
9. [J.M. Valverde](#), A. Shatillo, R. De Feo, O. Gröhn, A. Sierra, J. Tohka. **RatLesNetv2: A Fully Convolutional Network for Rodent Brain Lesion Segmentation.** In *Frontiers in Neuroscience*, 2020    

Submitted manuscripts

1. [J.M. Valverde](#), M. Østergaard, A. Rodriguez-Palomo, P. A. S. Vibe, N. K. Wittig, H. Birkedal, A. Dahl. **Disconnect to Connect: A Data Augmentation Method for Improving Topology Accuracy in Image Segmentation.** Submitted.  
2. K. Wilkinghoff, N. Madan, [J.M. Valverde](#), K. Nasrollahi, R.T. Ionescu, R. Wisniewski, T.B. Moeslund, W. Wang, Z.H. Tan. **Out of Context: Reliability in Multimodal Anomaly Detection Requires Contextual Inference.** Submitted. 
3. L. Dvořáková, R.A. Salo, **J.M. Valverde**, O. Narvaez, H. Laakso, J. Kyyriäinen, T. Zehnder, T. Mueggler, B. Künnecke, A. Sierra, O. Gröhn **RAFF4, magnetization transfer and diffusion tensor MRI in a genetic mouse model of demyelination and remyelination** Submitted to NMR in Biomedicine.

International abstracts

1. V. Imani, [J.M. Valverde](#), M. Prakash J.D. Lewis, O. Gröhn, J. Tohka. **Fluid Intelligence Classification Based On Cortical WM/GM Contrast, Cortical Thickness and Volumetry.** In *Organization for Human Brain Mapping*, 2020. 